Microsoft Elevate

CAPSTONE PROJECT

Hotel Reservation Analysis Dashboard

PRESENTED BY

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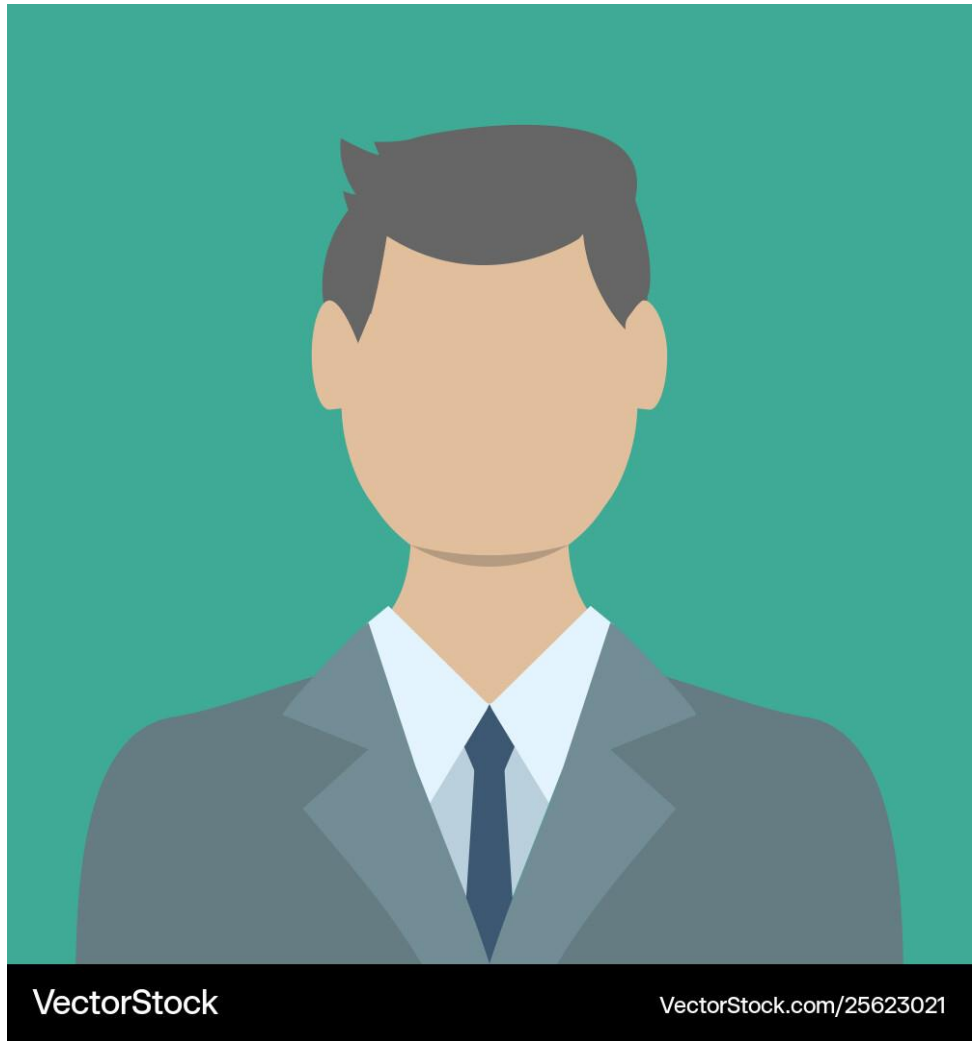


OUTLINE:

- Problem Statement
- Proposed System/Solution
- System Development Approach
- Algorithm & Deployment
- Result (Output Image)
- Conclusion
- Future Scope
- References

PROBLEM STATEMENT:

- Hotels generate large amounts of reservation data from different booking channels such as online platforms, corporate bookings, and direct reservations.
- However, many hotels struggle to effectively analyze this data to understand booking patterns, cancellation trends, and customer preferences.
- Without proper data analysis:
  - Hotels may face high cancellation rates.
  - Operational resources like rooms, parking, and meal plans may not be optimally utilized.
  - Opportunities for improving customer retention and satisfaction may be missed.
- Therefore, it is important to develop a system that analyzes hotel reservation data and provides meaningful insights to support better decision-making.

PROPOSED SOLUTION:

- The proposed solution is to develop an **interactive Power BI dashboard** that analyzes hotel reservation data and presents key insights to support hotel management decisions.
- The system includes the following components:
- **Data Collection**
  - Use the *Hotel.csv dataset from Kaggle*
  - Includes booking details such as lead time, room type, market segment, and special requests.
- **Data Preprocessing**
  - Clean and prepare the dataset
  - Handle missing values and ensure data consistency
  - Transform fields for better analysis
- **Data Visualization**
  - Build dashboards using Power BI
  - Create visual reports showing booking trends, cancellation rates, and operational insights.
- **Decision Support**
  - Provide insights to help reduce cancellations
  - Improve operational planning
  - Enhance customer experience.

SYSTEM APPROACH:

- **System Requirements**
  - Hardware:
    - Computer/Laptop
    - Minimum 8 GB RAM
  - Software:
    - Microsoft Power BI
    - Excel / CSV dataset
    - GitHub for project repository
- **Libraries / Tools Used**
  - Power BI Desktop
  - Data Visualization tools
  - Data transformation tools in Power BI
  - Kaggle dataset for hotel reservation data

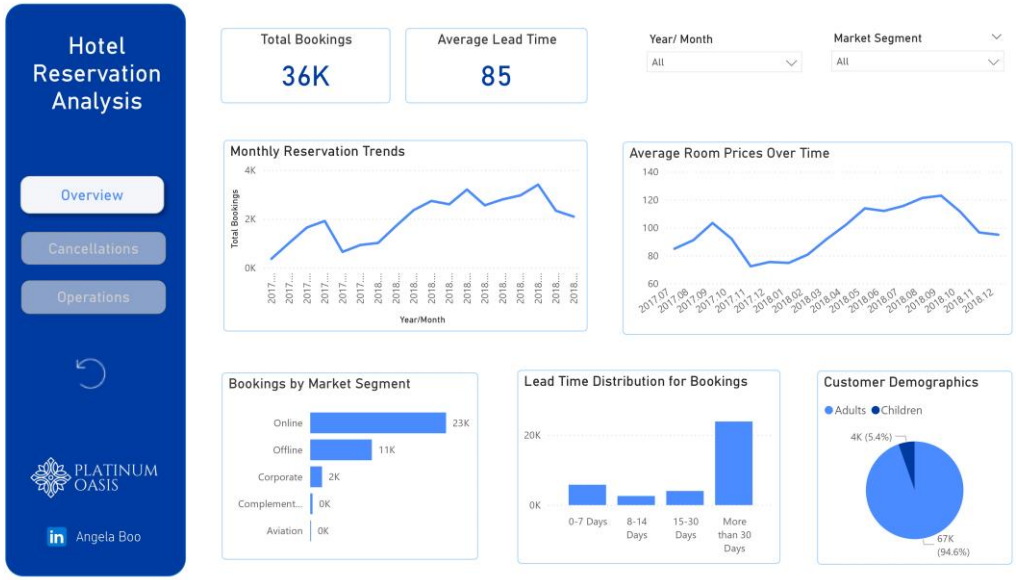
ALGORITHM & DEPLOYMENT:

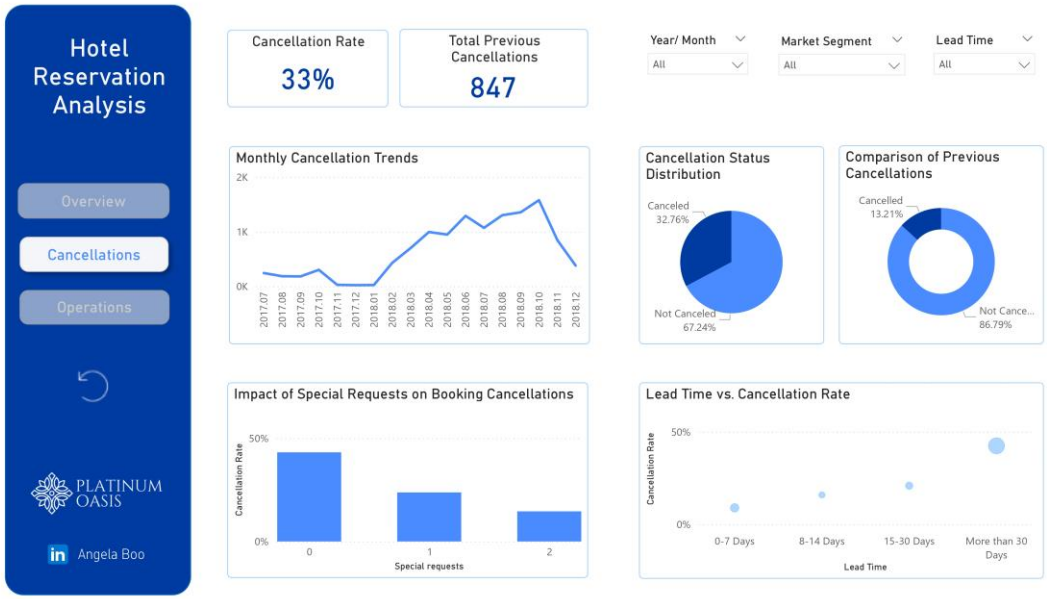
- Although this project focuses on **data analytics rather than predictive modeling**, the following analytical approach was used:
- **Data Input**
  - The system analyzes reservation data including:
  - Lead Time
  - Booking Status
  - Market Segment
  - Room Type
  - Special Requests
  - Repeated Guest indicator
- **Analysis Process**
  1. Import dataset into Power BI.
  2. Clean and transform data using Power Query.
  3. Create calculated measures and metrics.
  4. Develop visual dashboards to identify patterns.
- **Deployment**
  - The dashboard can be deployed:
  - On Power BI Service
  - Shared with hotel management
  - Used for real-time business insights.

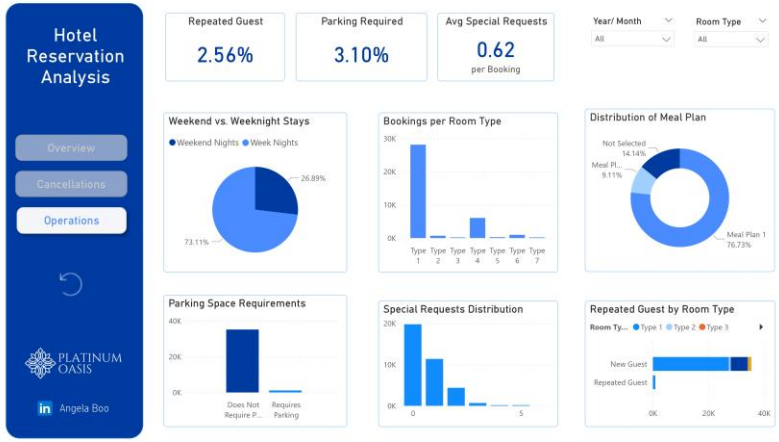
RESULT:

- The dashboard provides insights through three main pages:
- **Overview Dashboard**
  - Total bookings
  - Lead time analysis
  - Monthly booking trends
- **Cancellations Dashboard**
  - Cancellation rate by booking source
  - Impact of special requests on cancellations
  - Lead time vs cancellation analysis
- **Operational Insights Dashboard**
  - Most popular room types
  - Meal plan preferences
  - Parking space utilization
  - Returning guest analysis

RESULT:


Hotel Reservation Analysis
Overview
Cancellations
Operations
PLATINUM OASIS
Angela Boo
Total Bookings
36K
Average Lead Time
85
Year/ Month
All
Market Segment
All
Monthly Reservation Trends
Total Bookings
Year/Month
Average Room Prices Over Time
Bookings by Market Segment
Online 23K
Offline 11K
Corporate 2K
Complement... 0K
Aviation 0K
Lead Time Distribution for Bookings
20K
0K
0-7 Days
8-14 Days
15-30 Days
More than 30 Days
Customer Demographics
Adults
Children
4K (5.4%)
67K (94.6%)


Hotel Reservation Analysis
Overview
Cancellations
Operations
PLATINUM OASIS
Angela Boo
Cancellation Rate
33%
Total Previous Cancellations
847
Year/ Month
All
Market Segment
All
Lead Time
All
Monthly Cancellation Trends
Cancellation Status Distribution
Canceled 32.76%
Not Canceled 67.24%
Comparison of Previous Cancellations
Cancelled 13.21%
Not Cance... 86.79%
Impact of Special Requests on Booking Cancellations
Cancellation Rate
Special requests
Lead Time vs. Cancellation Rate
Lead Time
0-7 Days
8-14 Days
15-30 Days
More than 30 Days


Hotel Reservation Analysis
Overview
Cancellations
Operations
PLATINUM OASIS
Angela Boo
Repeated Guest
2.56%
Parking Required
3.10%
Avg Special Requests
0.62
per Booking
Year/ Month
All
Room Type
All
Weekend vs. Weeknight Stays
Weekend Nights
Week Nights
26.89%
73.11%
Bookings per Room Type
Distribution of Meal Plan
Not Selected 14.14%
Meal Pl... 9.11%
Meal Plan 1 76.73%
Parking Space Requirements
Does Not Require P...
Requires Parking
Special Requests Distribution
Repeated Guest by Room Type
Room Ty...
Type 1
Type 2
Type 3
New Guest
Repeated Guest

CONCLUSION:

- This project demonstrates how **data visualization and analytics can support decision-making in the hospitality industry**.
- Key outcomes include:
  - Identification of booking patterns and trends
  - Insights into cancellation behavior
  - Understanding customer preferences
  - Improved operational planning
- The Power BI dashboard enables hotel management to make **data-driven decisions**, improve service quality, and increase operational efficiency.

FUTURE SCOPE:

- Future improvements for this system may include:
  - Implementing **machine learning models to predict cancellations**
  - Integrating **real-time booking data**
  - Expanding the system to analyze **multiple hotel branches**
  - Adding **customer segmentation analysis**
  - Developing **automated reporting and alerts**
- These enhancements would further improve decision-making and operational efficiency.

REFERENCES:

- Kaggle Dataset:
  Hotel Reservation Dataset
- Power BI Documentation:
  https://learn.microsoft.com/power-bi/

GitHub Link: Link

Thank You